Sofiia Tretiak

AI Engineering and Model Evaluation | Python, Production Systems, PyTorch
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Portfolio: sofiia-tretiak.vercel.app

PROFESSIONAL SUMMARY

Medical Informatics engineer working in AI: I build systems that use models, and I evaluate what models produce. My own retrieval service runs in production and I maintain it end to end. I develop with AI coding assistants and test their output before it ships, and I work on model evaluation for an AI data platform. Earlier, a year in enterprise IT at Volvo on access management and integrations between enterprise systems, and research training GRU and Transformer models in PyTorch on EEG data.

PROFESSIONAL EXPERIENCE

Software Developer, AI Model Evaluation
Alignerr | Freelance, remote | Oct 2026 - Present

- Software development with AI coding assistants and structured evaluation of model output against written criteria. Project details under NDA

Software Engineer Intern, Digital Product Team
Volvo | Wroclaw, Poland | May 2025 - May 2026

- A year in the team responsible for the internal system that manages users, dealers and internal businesses, and for access management
- Administered accounts, access rights, licences and multi-factor authentication, and reworked the multi-factor authentication configuration workflow used across the organisation
- Tested REST API integrations between enterprise systems, logged bugs and improvement requests, and presented findings to stakeholders
- Investigated production incidents to root cause and verified data with SQL against production databases
- Wrote the instructions and operating procedures the team reused

Research Trainee, Neuroengineering Department
METAM, Karadeniz Technical University | Trabzon, Turkiye | Jun 2026 - Aug 2026

- Built the Python pipeline for EEG research data and trained GRU and Transformer models in PyTorch; ran ablation studies on data volume, channel count and signal length
- Found that the evaluation split placed recordings of the same stimulus in both training and test sets, rebuilt it, and reported the corrected result

PROJECTS

PharmaRAG - multilingual medicines assistant | personal project, live and open source | pharma-rag-nine.vercel.app

- FastAPI backend, React frontend, answers in six languages, 2-4 second response times
- Reads official documents from six national drug registers through APIs and files, then cleans, normalises and indexes about 400k chunks with hybrid re-ranking
- A citation behind every claim, general information marked as general, fixed responses for emergency questions, and automated tests on every change
- Wrong answers are traced through the retrieval layer to the source data instead of patched in the prompt

Technologies: Python, FastAPI, Qdrant, embeddings, Groq and Gemini APIs, React, Render, Vercel, pytest

Career Shark - Full-Stack Web Application | BEST Hackathon, Top 10

- Team lead of four at a 24-hour hackathon: React, FastAPI and MongoDB, built and deployed by the deadline, 7th of about 50 teams

Technologies: React, FastAPI, Python, MongoDB

TECHNICAL SKILLS

Python: my working language: FastAPI with pydantic models, pandas, NumPy, requests, pytest, PyTorch; REST APIs and JSON, request validation, error handling and retries, background jobs
AI Systems in Production: model APIs with automatic fallback between providers (Anthropic, Gemini, Groq), prompts for structured extraction, retrieval over large document collections with embeddings and hybrid re-ranking, citations traced back to source documents, answer safety tiers
Model Evaluation: structured review of model output against written criteria: hallucinations, inconsistencies and systematic failure patterns, with documented reasoning for every judgement; defining what counts as a correct answer before measuring it; data splitting and leakage
Machine Learning: deep learning in PyTorch (GRU, Transformer, CNN, ResNet), training, evaluation and ablation studies, data splitting and leakage, choosing metrics and knowing what they measure
Engineering Practice: Git with branching, pull requests and code review; automated unit and integration tests on every change; Docker; cloud deployment (Render, Vercel); Linux command line; Azure DevOps, Jira, ServiceNow
Data and Front End: SQL queries and prepared views against production databases; ingestion from APIs and files, cleaning and deduplication at scale; Qdrant and vector search; MongoDB; React, JavaScript, HTML and CSS

EDUCATION AND CERTIFICATIONS

Master of Science, Medical Engineering and eHealth
UAS Technikum Wien, Vienna | 2026 - 2028

- Enrolled full-time in Vienna, expected graduation 2028

Bachelor of Engineering, Medical Informatics and Bioengineering
Wroclaw University of Science and Technology | 2022 - 2026

- Specialisation in Medical Informatics, taught entirely in English: programming, algorithms and data structures, databases and SQL, software engineering, artificial intelligence 1 and 2, digital signal processing, computer networks, statistics (highest grade of the degree)

LANGUAGES

Ukrainian (native), English (C1), Polish (C1), German (A2, actively learning)